

CAT Algebra – Mixed Practice Set 3 of 3

Topics: Polynomials · Quadratics · Binomial · Inequalities · Functions · GIF · Modulus · Logarithms · Progressions · Co-ordinate Geometry · Maxima & Minima

QUESTIONS

Q1.

[Maxima & Minima]

$x^2+ax+1=0$ has both roots real and negative. Minimum possible value of a :

- (1) 0
- (2) 1
- (3) 2
- (4) 3

Q2.

[Maxima & Minima]

x, y, z real with $x+y+z=5$, $xy+yz+xz=3$. Largest value x can have? (CAT 2002)

- (1) $3+\sqrt{13}$
- (2) $\sqrt{19}$
- (3) $13/3$
- (4) None of these

Q3.

[Maxima & Minima]

a, b, c, d positive reals, $abcd=1$. Minimum of $(1+a)(1+b)(1+c)(1+d)$? (CAT 2001)

- (1) 4
- (2) 1
- (3) 16
- (4) 18

Q4.

[Maxima & Minima]

$f(x)=\max(2x+1, 3-4x)$. Minimum value of $f(x)$? (CAT 2006)

- (1) $1/3$
- (2) $1/2$
- (3) $2/3$
- (4) $4/3$
- (5) $5/3$

Q5.

[Maxima & Minima]

$T=\{3, 11, 19, \dots, 467\}$. Max elements in $S \subseteq T$ with no two summing to 470? (CAT 2003)

- (1) 32
- (2) 28
- (3) 29
- (4) 30

Q6.

[Co-ordinate Geometry]

Condition for $x+2y-9=0$, $3x+5y-5=0$ and $ax+by-1=0$ to be concurrent:

- (1) $35a+22b=1$
- (2) $35a-22b=-1$
- (3) $35a+22b=0$
- (4) $22a+35b=-1$

Q7.

[Co-ordinate Geometry]

Perpendicular distance between $3x+4y+3=0$ and $3x+4y+12=0$:

- (1) $11/5$
- (2) 2
- (3) 3
- (4) $9/5$

Q8.

[Co-ordinate Geometry]

Points $(-a,-b)$, $(0,0)$, (a,b) and (a^2,ab) are:

- (1) Vertices of parallelogram
- (2) Vertices of rectangle
- (3) Collinear
- (4) On a circle

Q9.

[Co-ordinate Geometry]

Circle 1: centre $A(5,3)$ radius 12. Circle 2: centre $B(2,-1)$ radius 6. Shortest distance between them:

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q10.

[Graphs]

How many straight lines through $(0,30)$ have a negative integral slope?

- (1) 1
- (2) 8
- (3) 16
- (4) Infinite

ANSWER KEY

Q1: (3) 2 **Q2:** (3) $13/3$ **Q3:** (3) 16 **Q4:** (5) $5/3$ **Q5:** (4) 30

Q6: (2) $35a-22b=-1$ **Q7:** (4) $9/5$ units **Q8:** (3) Collinear **Q9:** (1) 1 **Q10:** (2) 8

DETAILED SOLUTIONS

Solution Q1 — [Maxima & Minima]

Answer: (3) 2

Real roots: $a \geq 2$ or $a \leq -2$. Both negative: $a > 0$. So minimum $a=2$.

Solution Q2 — [Maxima & Minima]

Answer: (3) $13/3$

Quadratic in y with non-negative discriminant gives $x \leq 13/3$. Maximum is $13/3$.

Solution Q3 — [Maxima & Minima]

Answer: (3) 16

By AM-GM: product $\geq 16 \cdot (abcd)^{1/2} = 16$. Equality at $a=b=c=d=1$.

Solution Q4 — [Maxima & Minima]

Answer: (5) $5/3$

Lines intersect at $x=1/3$. $f(1/3)=2(1/3)+1=5/3$ (minimum).

Solution Q5 — [Maxima & Minima]

Answer: (4) 30

T has 59 elements. 29 complementary pairs + middle element 235. Pick one from each pair + 235 = 30.

Solution Q6 — [Co-ordinate Geometry]

Answer: (2) $35a-22b=-1$

Solving first two: $y=22$, $x=-35$. Substituting: $-35a+22b=1 \rightarrow 35a-22b=-1$.

Solution Q7 — [Co-ordinate Geometry]

Answer: (4) $9/5$ units

Distance = $|12-3|/\sqrt{9+16} = 9/5$ units.

Solution Q8 — [Co-ordinate Geometry]

Answer: (3) Collinear

All lie on $y=(b/a)x$. All four are collinear.

Solution Q9 — [Co-ordinate Geometry]

Answer: (1) 1

$AB=5 < 6$, so circle 2 is inside circle 1. Shortest distance = $12-(6+5)=1$.

Solution Q10 — [Graphs]

Answer: (2) 8

Slope = $-30/p$ where p divides 30. Divisors: 1, 2, 3, 5, 6, 10, 15, 30 $\rightarrow$ 8 lines.
